NISSAN NORTH AMERICA: AIR HANDLING UNITS CONTROL UPGRADE DELIVERS MASSIVE ENERGY SAVINGS

SECTOR
Industrial

LOCATION
Canton

PROJECT SIZE
4.7 million Square Feet

FINANCIAL OVERVIEW
Project Cost: $252,000

BACKGROUND

Nissan's Canton, Mississippi plant is one of four of the company's manufacturing facilities in the United States. Opened in 2003, the Canton plant is a 4.5 million square foot plant that can produce up to 410,000 vehicles annually. After completing a chilled water optimization project, Nissan's energy team shifted their focus to the plant's air handling units (AHUs), which rely on operator inputs to circulate air to different parts of the plant. In 2019, Nissan evaluated an AHU upgrade that included automated controls to manage airflow without ongoing plant operator management. To test out the solution, six AHUs were upgraded with the new controls and significant energy reductions were realized. Nissan approved additional funding for the project and expanded the scope to upgrade the remaining AHUs with new controls at the Canton plant. The project has been a tremendous success, saving over 4.8 million kilowatt-hours (kWh) annually.

During the initial 2019 AHU study, Nissan partnered with a third-party controls vendor to implement the upgrade on six AHUs. Nissan evaluated the effectiveness of the upgrade by monitoring the energy usage for two days after implementation. Then, they turned off the controls system for another two days to compare the difference in energy usage. The difference in AHU system load was measured to be 24 kilowatts (kW) per AHU. Based on the success of the project, the Canton plant upgraded 35 additional AHUs with the new controls.

SOLUTIONS

After proving out the controls solution with six AHUs, Nissan funded a project to upgrade 35 more AHUs throughout the trim shop area of the plant. The 35 AHUs deliver over 2.1 million cubic feet per minute of conditioned air to maintain temperature, humidity, and air quality. The controls system will lower the speed of the AHU motors once set points are met. This enables the temperature and humidity to be maintained while running the motors at a lower kilowatt (kW) load. Additionally, the controls include CO_2 sensors to monitor air quality and adjust outdoor air ventilation accordingly. If the air quality is low based on higher CO_2 levels, the dampers modulate open to bring in more outside air.

The AHU controls upgrade lowered the power demand of the 35 AHUs resulting in 48(kW of total load shed. The project is estimated to reduce the energy usage of the plan by over 4.8 million kWh or the equivalent of 1,800 metric tons of CO_2 emissions. The success of the project has led Nissan to invest in 30 additional AHU upgrades at the Canton plant. By the end of 2021, the 30 AHU controls will be upgraded, and the company anticipates additional savings of 4.1 million kWh annually and over 1,500 metric tons of CO_2.

OTHER BENEFITS

The automation and controls upgrade to Nissan's AHUs ensure more consistent operations are maintained, which will lead to greater energy efficiency, lower operating costs, and improved reliability. Plant operators are now freed up from the time and effort required to operate the AHUs since the controls now monitor the interior and outdoor temperatures to reduce cooling energy needs and improve comfort levels throughout the plant. Looking forward, Nissan plans to replicate the success of this project and is in the process of implementing the same AHU controls solution at two other facilities—Nissan's U.S. headquarters in Tennessee and vehicle test facility in Arizona.